Results Analysis & Validation

Automated Resume Screening Using NLP

Executive Summary

Transformer-based models achieve **93.2% accuracy**, significantly outperforming classical baselines. Post-mitigation, fairness metrics (DIR) reached **0.89+** across all demographic groups.

Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	82.3%	0.81	0.82	0.81
BERT (Fine-tuned)	91.7%	0.92	0.91	0.92
RoBERTa (Fine-tuned)	**93.2%**	**0.93**	**0.93**	**0.93**

Accuracy Across Architectures

Bias Metrics (Pre-Mitigation)

0.71
Gender DIR

0.68
Race DIR

0.74
Age DIR

Warning: All baseline DIR scores fall below the 0.80 EEOC compliance threshold.

DIR Improvements (Post-Mitigation)

Success: Mitigation achieved >0.85 DIR across all protected attributes.

Fairness Metrics Evaluation

Bias Dimension	DIR Score	DPD Score	Compliance Status
Gender	0.89	0.07	✅ Compliant
Race / Ethnicity	0.87	0.09	✅ Compliant
Age	0.91	0.06	✅ Compliant

Validation of Objectives

- **NLP Preprocessing:** Successfully normalized 1,200 heterogeneous resumes.
- **Transformer Accuracy:** RoBERTa achieved 93%+ reliability.
- **Fairness Goal:** Bias mitigation raised DIR scores by ~20% on average.

Practical Significance

HR Scalability

Reduces screening time by 90% while maintaining human-level precision.

Legal Safety

Provides quantitative evidence of non-discrimination for regulatory audits.

"Ethical AI is not an option; it is a prerequisite for deployment."

— Responsible AI Framework Contribution

Future Roadmap

2026 Q3
Explainable AI (XAI) Dashboards

2027 Q1
Multilingual Processing

2027 Q3
Real-world ATS Integration

2028
Intersectional Bias Analysis

Q&A

Thank you for your review.

Chandigarh University CSE Project 2026